What is the Lazy regime ?

Learning with training loss going to 0, but params hardly changes.

so lets take $\theta(0)$ at init

$\theta(T)$ for when loss $= 0$

$\rightarrow \hat{y} = f(\theta(T))$

first order taylor expansion

$$f(x) \approx f(a) + (x-a)\nabla_x f(x)\Big|_{x=a}$$

$$\hookrightarrow \hat{y} \approx f(\theta(0)) + (\theta(T) - \theta(0))\nabla_\theta f(\theta(0))$$

thus $\Delta_\theta = \theta(T) - \theta(0)$

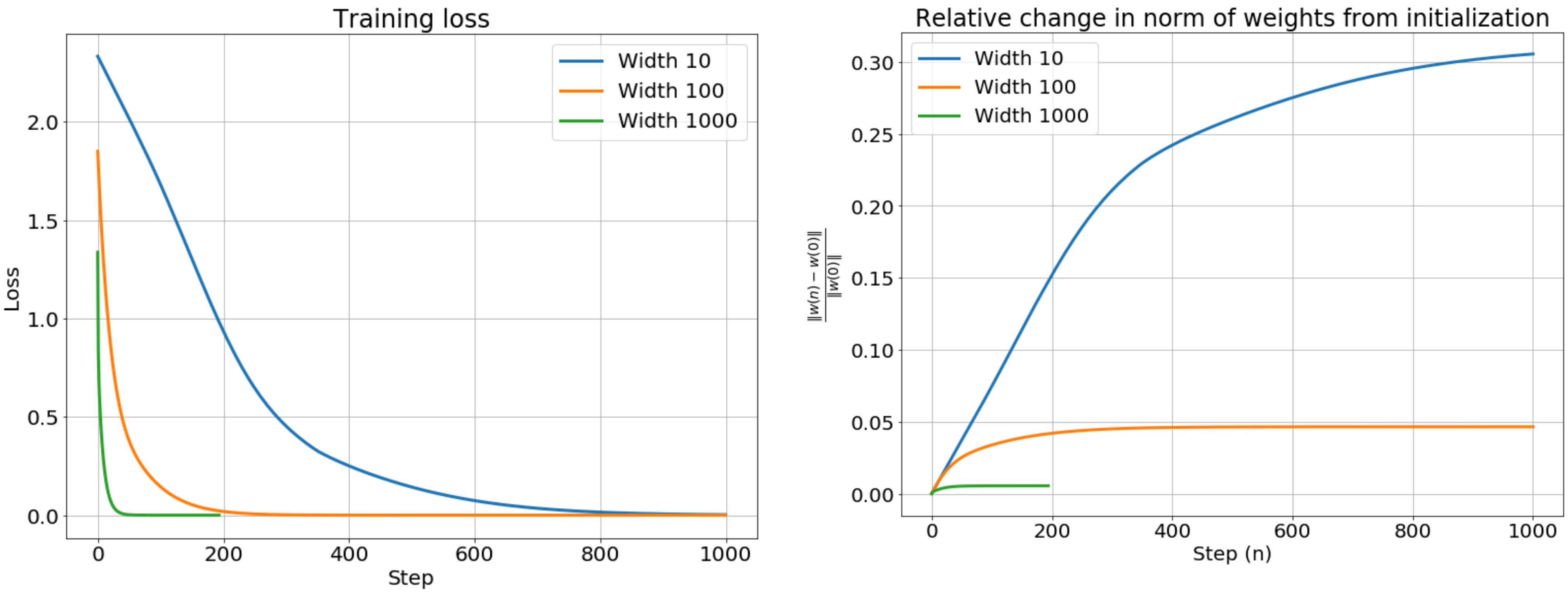
$$= \frac{\|\hat{y} - f\ \theta(0)\|}{\|\nabla_\theta f(\theta(0))\|}$$

$$\nabla_\theta f(\theta(T)) \approx \nabla_\theta f(\theta(0)) + \nabla_\theta^2 f(\theta(0))\Delta\theta$$

$$= \nabla_\theta f(\theta(0)) + \nabla_\theta^2 f(\theta(0)) \frac{\|\hat{y} - f(\theta(0))\|}{\|\nabla_\theta f(\theta(0))\|}$$

Ainsi $\Delta(\nabla_\theta f) = \nabla_\theta f(\theta(T)) - \nabla_\theta f(\theta(0))$

$$= \nabla_\theta^2 f(\theta(0)) \frac{\|\hat{y} - f(\theta(0))\|}{\|\nabla_\theta f(\theta(0))\|}$$



when overparametrized NN goes to infinite width, we have little to no changes in the weights over time, but learning error goes to 0. This is *lazy*.

Since w almost doesnt change, we will taylor dev it.

$f(x, w^\tau) = f(x, w^0) + \nabla_w f(x, w^0)(w^\tau - w^0) \Rightarrow y(x^\tau) = y(w^0) + \nabla_w y(w^0) \triangle_w$ we see that it becomes a linear model in w, so linear regression for w.

$$\text{so} \quad \Delta_w = (w^{(\tau)} - w^{(0)})$$

$$= \frac{\| y(w^{(\tau)}) - y(w^{(0)}) \|}{\| \nabla_{w^{(0)}} y(w^{(0)}) \|}$$

distance moved in w space

$$\nabla_w f(w^{(\tau)}) \approx \nabla_w f(w(0)) + \nabla_w^2 f(w(0)) \Delta_w$$

$$= \nabla f(w(0)) + \nabla_w^2 f(w(0)) \frac{\| \hat{y} - f(w(0)) \|}{\| \nabla_w f(w(0)) \|}$$

$$\text{Ainsi} \quad \Delta(\nabla_w f) = \nabla_w f(w(\tau)) - \nabla_w f(w(0))$$

$$= \nabla_w^2 f(w(0)) \frac{\| y - f(w(0)) \|}{\| \nabla f(w(0)) \|}$$

$$K(w) = \frac{\Delta(\nabla_w y)}{\nabla_w y(w^{(0)})} = \| y(w^{(\tau)}) - y(w^{(0)}) \| \frac{\nabla_w^2 y(w^{(0)})}{\| \nabla_w y(w^{(0)}) \|^2}$$

the amount of change in w required to produce a change of $\| y(w^{(\tau)}) - y(w^{(0)}) \|$ in y causes a negligible change in the Jacobian $\nabla_w y(w)$

Do it by init

$$K(w) = \| y(w^{(t)}) - y(w^{(0)}) \| \frac{\nabla_w^2 \, y(w^{(0)})}{\| \nabla_w \, y(w^{(0)}) \|^2}$$

assume 0 at init i.e. $y(w^{(0)}) = 0$

and scale the output of the model by α

$$\rightarrow K_\alpha(w) \approx \frac{\alpha \nabla_w^2 \, y(w^{(0)})}{\| \nabla_w \alpha y(w^{(0)}) \|^2}$$

$$= \frac{\alpha}{\alpha^2} \frac{\nabla_w^2 \, y(w^{(0)})}{\| \nabla_w \, y(w^{(0)}) \|^2}$$

$$= \frac{1}{\alpha} \frac{\nabla_w^2 \, y(w^{(0)})}{\| \nabla_w \, y(w^{(0)}) \|^2}$$

so $K_\alpha(w) \longrightarrow 0$
$\alpha \longrightarrow \infty$

INVERSE RELATIVE SCALE of the model

GRADIENT FLOW (learning dynamics).

$$w^{(t+1)} = w^{(t)} - \eta \nabla \mathcal{L}_w(w^{(t)})$$

$$\Longrightarrow \frac{w^{(t+1)} - w^{(t)}}{\eta} = -\nabla \mathcal{L}_w(w^{(t)})$$

we can say

$$\frac{dw}{dt} = -\nabla \mathcal{L}_w(w^{(t)})$$

$$\mathcal{L}_w(w^{(t)}) = \frac{1}{N} \sum^{N} \ell(y(w^{(t)}), x)$$

$$\Longrightarrow \nabla_w \mathcal{L}(w^{(t)}) = \frac{1}{N} \sum^{N} \nabla_w y \, \nabla_y \ell(y, x)$$

donc $\dfrac{dw}{dt} = -\dfrac{1}{N} \sum^{N} \nabla_w y \, \nabla_y \ell(y, x)$

donc $\dfrac{dy}{dt} = \dfrac{dy}{dw} \dfrac{dw}{dt}$

$$\Longrightarrow \frac{dy}{dt} = -\frac{1}{N} \sum^{N} \underbrace{\nabla_w y^T \nabla_w y}_{NTK} \nabla_y \ell(y, x)$$

so if we are close to linear regression ($\kappa \ll 1$) then the Jacobian matrix of the model outputs does not change as training progresses, i.e:

$$\nabla y(w^{(t)}) \approx \nabla y(w^{(0)}) \Longrightarrow NTK(y(w^{(t)})) = NTK(y(w^{(0)}))$$

that's why we call it kernel regime: bc kernel stays the same during the whole training

$$\frac{dy(w^{(t)})}{dt} = -NTK(y(w^{(0)}))(y(w^{(t)}) - \bar{y})$$

$\hookrightarrow$ ODE

$$\dot{u} = -NTK\, y(w^{(0)})\, u$$

$$u = (y(w^{(t)}) - \bar{y})$$

$y(w^{(t)}) = \bar{y}$ equilibrium (training loss $= 0$).